Generative AI Analysis and Ethics Report

A Convolutional Variational Autoencoder for Fashion-MNIST Clothing Images

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1. Overview

The goal of this project is to **generate new, realistic clothing images (28 x 28 grayscale) and to learn a smooth latent space of clothing styles**. I implemented a convolutional Variational Autoencoder (VAE) in PyTorch and trained it on Fashion-MNIST, a public dataset of 70,000 real Zalando product photos in ten clothing classes (Xiao et al., 2017). A VAE is a latent-variable generative model with two parts: an encoder that maps an image to a Gaussian distribution over a low-dimensional code, and a decoder that maps a code back to an image. Both parts are trained together by maximising a lower bound on the data log-likelihood, the evidence lower bound or ELBO (Kingma & Welling, 2014).

I chose a VAE over a GAN for three reasons. It trains by minimising a single loss, so there is no adversarial game that can become unstable. It gives an explicit latent space, which I need for interpolation. And the same model can both reconstruct images, which I can measure, and generate new images from the prior. The trained model has 1,688,289 parameters and a 16-dimensional latent space, and it trained in 2.4 minutes on a CPU. This report describes the data, the design and training, the quality and failure modes of the outputs, and the ethical questions this specific system raises.

2. Dataset or Prompt Description

Source and licence. Fashion-MNIST was released by Zalando Research and is available at https://github.com/zalandoresearch/fashion-mnist under the MIT License. It contains front-view product images from Zalando's online catalogue that the authors converted to 28 x 28 grayscale. It was designed as a harder drop-in replacement for the MNIST digit benchmark (Xiao et al., 2017). The images are real photos of real products. They are not synthetic and not AI-generated. I use them only for this academic project. The notebook downloads the data automatically with `torchvision.datasets.FashionMNIST(root='./data', download=True)`, so anyone can rerun the project without an account or manual steps.

Size and structure. The dataset has 60,000 training and 10,000 test images in ten classes: T-shirt/top, trouser, pullover, dress, coat, sandal, shirt, sneaker, bag and ankle boot. Each class has exactly 6,000 training and 1,000 test images, so the classes are perfectly balanced. Pixels are 8-bit integers (0–255). I held out 5,000 random training images (fixed seed) for validation, which leaves 55,000 for training. The 10,000 test images are used only for the final evaluation.

How it supports the task. The items are centred on a black background with a consistent pose. The model can therefore spend its capacity on garment shape and style instead of on background or

position, which makes the dataset well suited to learning a compact latent space of clothing styles. Inspecting the samples (Figure 1) also showed in advance where problems were likely. Trousers and footwear have distinctive outlines, but T-shirts, pullovers, coats and shirts share almost the same silhouette. Many items carry fine detail, such as a printed "89" T-shirt, a plaid shirt and striped sneakers. For preprocessing, I scaled pixels to [0, 1] with `ToTensor()`, because the binary cross-entropy loss expects targets in that range. I did not use augmentation, because flips or rotations would create poses that do not occur in the catalogue.

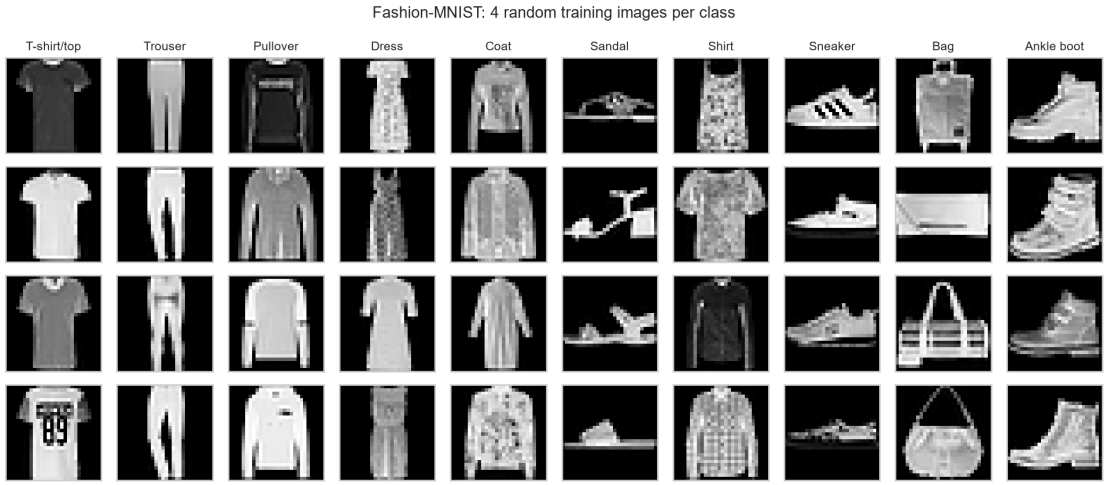


Figure 1. Four random training images per class, displayed before modelling.

3. Model Design and Training Approach

Architecture

The encoder has two stride-2 convolutions ($1 \to 32 \to 64$ channels, 4 x 4 kernels, ReLU) that reduce the image from 28 x 28 to 14 x 14 and then 7 x 7. These are followed by a 256-unit fully connected layer and two linear heads that output the mean μ and log-variance $\log \sigma^2$ of a 16-dimensional Gaussian. A latent code is sampled with the reparameterisation trick, $z = \mu + \sigma \times \varepsilon$ with $\varepsilon \sim N(0, I)$ (element-wise product). This makes the sampling step differentiable, so the encoder can be trained by backpropagation (Kingma & Welling, 2014). The decoder mirrors the encoder: two fully connected layers, a reshape to 64 x 7 x 7, and two stride-2 transposed convolutions back to 1 x 28 x 28, with a sigmoid so that every output pixel is a Bernoulli mean in [0, 1]. I used convolutions rather than a fully connected network because they share weights across positions and capture local structure such as collars, straps and soles.

Loss, optimiser and hyperparameters

The loss is the negative ELBO for each image: the binary cross-entropy between input and output, summed over the 784 pixels (the negative log-likelihood of a Bernoulli decoder), plus the closed-form KL divergence between the encoder distribution and the standard normal prior (Kingma & Welling, 2014). The KL term is what makes random codes from N(0, I) decode into realistic images. Higgins et al. (2017) show that weighting this term with a factor β trades reconstruction detail for a smoother, more regular latent space. I used $\beta = 1$ for the main model and tested $\beta = 4$ as a single controlled

change. The training settings were Adam with learning rate 0.001, batch size 128 (430 updates per epoch), and 25 epochs, with the seed fixed for every random number generator. Adam adapts the step size for each parameter, similar to the adaptive optimiser used in the original VAE work (Kingma & Welling, 2014), and it handles the very different gradient scales of the reconstruction and KL terms.

Table 1. Main design choices.

Setting	Value	Reason
Latent size	16	Room for shape, sleeve length, heel and brightness; still smooth enough to sample
Reconstruction loss	BCE, summed over pixels	Bernoulli likelihood for [0, 1] pixels
Regulariser	KL to N(0, I), $\beta = 1$	Makes prior samples decodable
Optimiser	Adam, lr = 0.001	Stable adaptive steps
Batch / epochs	128 / 25	Converged within the CPU budget (2.4 min)

Justification from my training results

Figure 2 shows that training was stable. The validation loss fell from 254.38 to 236.40 nats per image and improved by only 0.21 nats over the last five epochs, so 25 epochs were enough and I did not need to shorten training. At the end, the validation loss was only 1.10 nats above the training loss, so the model did not over-fit. The curves also show the expected tension between the two loss terms. Reconstruction loss fell (validation 241.55 → 221.60) while the KL term rose (12.83 → 14.80): the encoder put more information into z because this bought better reconstructions. A per-dimension KL diagnostic showed one training challenge: only 9 of the 16 latent dimensions carry information, and the rest stay at the prior. Because the KL term charges a cost for every dimension the encoder uses, the model leaves dimensions it does not need at the prior. So 16 dimensions were more than this model needed. This did not prevent good samples, but it shows that the latent size could be reduced.

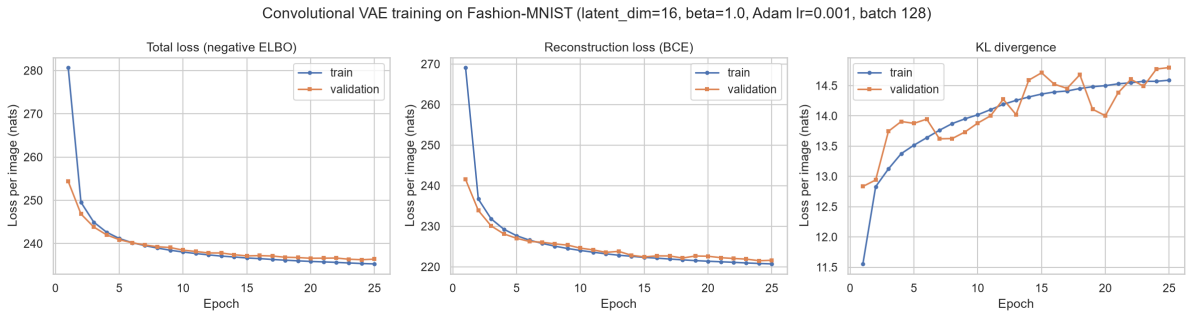


Figure 2. Training and validation loss per epoch: total negative ELBO (left), BCE reconstruction (middle) and KL divergence (right), in nats per image.

4. Output Evaluation and Interpretation

Quantitative overview. On the 10,000 test images the model reaches a negative ELBO of 237.83 nats per image, a reconstruction BCE of 219.22 nats and a mean squared reconstruction error of 0.0124 per pixel. These numbers measure reconstruction, not the realism of new samples. I therefore based my evaluation of generated outputs mainly on a structured visual inspection of 64 random samples (Figure

3), 16 reconstructions (Figure 4) and three interpolations (Figure 5). All positions below refer to the labelled grids.

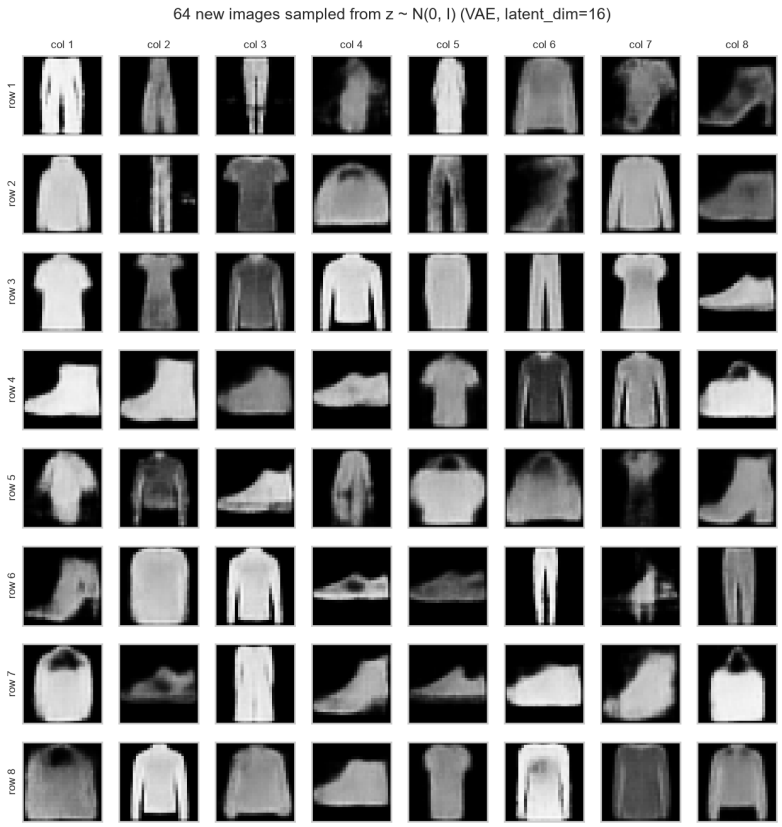


Figure 3. 64 new images decoded from z ~ N(0, I). Rows and columns are numbered so that individual samples can be referenced.

Quality and variability of new samples

Most samples are recognisable garments with clean outlines. Examples are the wide white trousers at row 1, column 1, the slim trousers at row 6, column 6, a sneaker with a clear sole line at row 3, column 8, two flat ankle boots at row 4, columns 1–2, and a bag with a handle at row 7, column 8. To check diversity, I labelled each sample with the class of its nearest training image. The 64 samples cover 10 of 10 classes, with coats (10), sneakers (9) and ankle boots (9) the most frequent. Samples also vary within a class, for example wide vs slim trousers and heeled boots (row 1, column 8; row 5, column 8) vs flat boots (row 4, column 1). The model has therefore not collapsed to a few prototypes.

Failure cases (concrete examples)

- **Blur and lost texture.** None of the 64 samples has a print, logo, stripe or check pattern, although the training data contains many (Figure 1). All surfaces are smooth grey gradients. The reconstructions show the same effect most clearly. The pullover #3 carries a large "Lee" logo; its reconstruction keeps the pullover's shape but erases the logo completely (MSE 0.030, the worst of the 16). The plaid shirt #15 becomes a plain grey shirt (MSE 0.022). Plain items such as trouser #2 and dress #4 are almost perfect (MSE 0.003). Because the decoder outputs the mean value of each pixel,

uncertain detail is averaged into grey.

- **Class blending.** The sample at row 1, column 7 joins a T-shirt-like top to a boot-like lower outline. Row 6, column 2 is a sleeveless rounded rectangle halfway between a bag and a pullover. Row 1, column 4 is a shapeless grey blob that matches no class.
- **Artefacts.** Row 6, column 7 is a fragmented shape with horizontal streaks. Row 8, column 6 is a white pullover with a dark smudge where a printed graphic should be.
- **Weak classes.** Only 1 of the 64 samples resembles a sandal (row 7, column 2), and it is a closed, dark shoe without straps. Across the full test set, the highest per-class MSE belongs to sandal (0.0193) and bag (0.0182), and the lowest to trousers (0.0062). Thin straps and varied bag shapes are hard to reproduce pixel by pixel. Ranked by BCE, pullover (283.7) and shirt (278.3) are worst. In the latent space, shirts, pullovers and coats are the most mixed classes: only 44%, 53% and 55% of their ten nearest latent neighbours share their class, against 89% for trousers. This overlap is the likely cause of the blending failures.

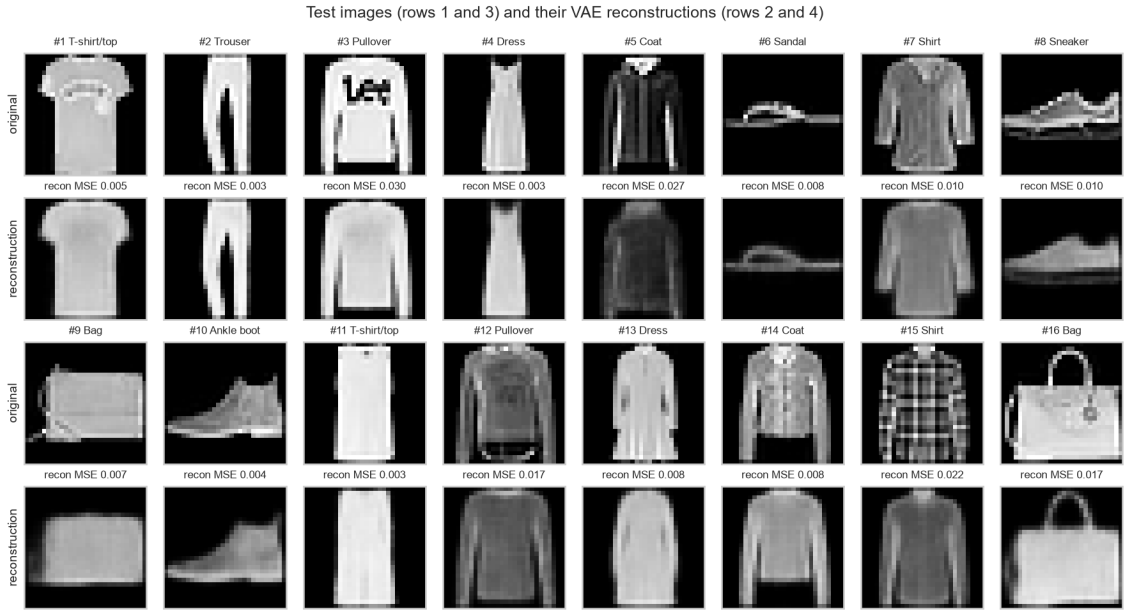


Figure 4. Test images (rows 1 and 3) and their reconstructions (rows 2 and 4), with per-image MSE.

What the interpolations show

Figure 5 decodes straight lines between the latent means of two real test images. Within the footwear family the latent space is smooth and meaningful: from sneaker to ankle boot the sole stays fixed while the shaft rises steadily, and every frame is a plausible shoe. Between families the path passes through other classes. The shirt-to-trouser path goes through a sleeveless, dress-like shape ($a \approx 0.33$–0.44) before two legs separate. The sandal-to-bag path first turns the open sandal into a closed sneaker-like shoe and only then into a box-shaped bag without the original bag's handle. These middle frames are hybrids that do not exist as real products, so the space is smooth but not organised by class.

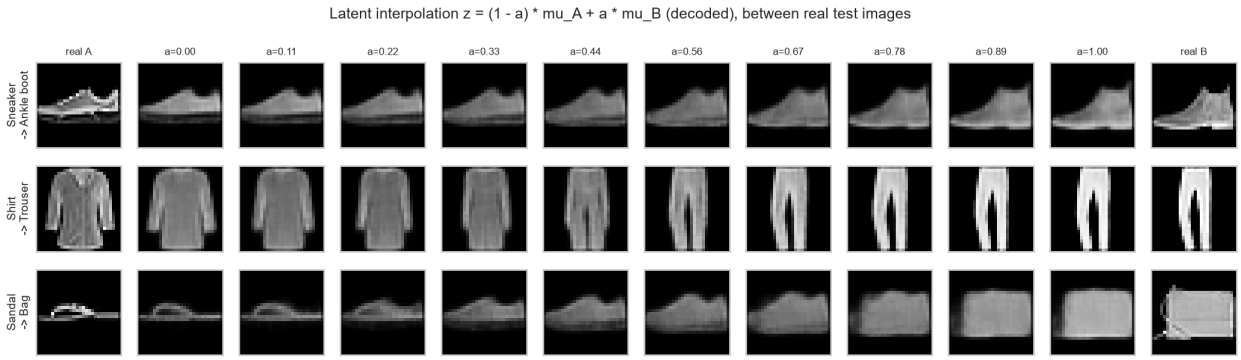


Figure 5. Latent interpolation for three class pairs. The real start and end images are shown in the outer columns.

Effect of the KL weight ($\beta = 4$)

With everything else fixed, raising β to 4 lowered the test KL from 14.76 to 6.78 nats and reduced the active latent dimensions from 9 to 6. Test MSE rose by 42% (0.0124 $\rightarrow$ 0.0176), and the samples were visibly blurrier, with faint, ghost-like shoes. This confirms the trade-off described by Higgins et al. (2017). For my goal of realistic samples, $\beta = 1$ is the better setting.

5. Ethical Considerations and Responsible Use

(a) Bias and representation

Everything this model can produce comes from one retailer's catalogue. Fashion-MNIST was built from Zalando product images (Xiao et al., 2017), which mostly show mainstream, Western-market clothing photographed on a plain background. Its ten categories are retail labels that follow gender-coded shop conventions (for example, dresses form their own class, while the dataset itself has no gender or body-type information). My outputs show this limit directly. Every sample in Figure 3 falls into one of the ten catalogue classes. The latent space has no region for garments outside them, such as a sari, a kimono, a hijab or adaptive clothing. Interpolations between classes produce hybrids rather than new styles. If a system like this were used to propose designs or to fill product databases, it would reproduce one retailer's view of what clothing is, and it would quietly leave out other cultures and body types. A responsible claim about this model is therefore narrow: it represents the Fashion-MNIST catalogue, not clothing in general.

(b) Creative ownership and memorisation

The training images show products designed by other people and photographed by Zalando, so a generator that reproduced them would copy their work. Carlini et al. (2023) showed that large image generators, including diffusion models, can memorise and regenerate individual training images. They measured this as a very small pixel-space distance between a generated image and a training image. I applied the same idea at small scale. For all 64 samples, the mean L2 distance to the nearest of the 60,000 training images was 3.35 (minimum 1.87). For 64 real, unseen test photos it was 3.34 (minimum 1.67). My samples are therefore not closer to the training data than genuinely different real products are. Figure 6 shows that the nearest neighbours share only a rough outline; the heeled boot at row 1, column 8, for example, is matched to a high-top sneaker with a bright logo. With about 1.7 million

parameters and a 16-dimensional bottleneck, the model can keep only coarse shape, and that limits copying. Pixel distance is still a crude test. Blurry average images can be close to many training images without copying any of them, so this check supports, but does not prove, the absence of memorisation.

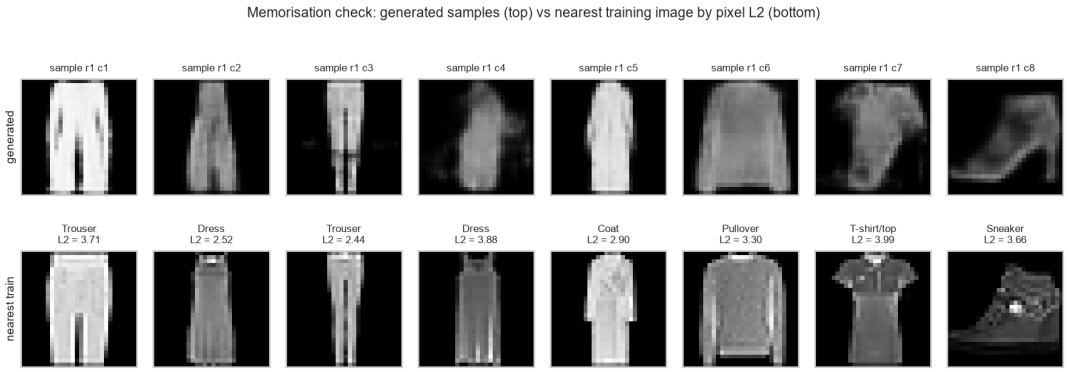


Figure 6. Memorisation check: the first eight generated samples (top) and their nearest training images by pixel L2 distance (bottom).

(c) Misuse

The most realistic misuse of clothing generators is synthetic product photos for fake or misleading online listings, such as selling items that do not exist or imitating a brand's catalogue. My model is a low risk here. Its outputs are 28 x 28, grayscale, visibly blurry and without logos (Figures 3 and 4), so nobody would mistake them for real product photos. The risk grows quickly with scale, however. High-resolution colour models trained on scraped catalogues, such as modern diffusion models, can produce convincing product images and can reproduce specific training images (Carlini et al., 2023). The same design would need much stronger safeguards at that scale.

How my design choices relate to responsible use

- **Documented, licensed, real data:** I used a public dataset with a clear MIT licence and documented origin, download method and limits, instead of scraping product photos whose owners did not agree to this use.
- **Memorisation check as part of evaluation:** the nearest-neighbour test is built into the notebook, so any retrained or larger version is checked for copying before its outputs are used.
- **Low resolution and academic scope:** the model's size and output resolution fit an academic study of generative modelling. I do not release it as a product-image generator, and any generated image shown outside this project should be labelled as synthetic.
- **Honest reporting of limits:** by reporting per-class errors and failure cases, I make it visible which items the model represents poorly (sandals, bags, detailed tops) rather than presenting it as uniformly good.

6. Limitations and Future Improvements

- **Blurry outputs.** With a pixel-wise Bernoulli likelihood, the decoder predicts the average of plausible pixel values. This erased logos, prints and plaid in my results (MSE 0.030 for the logo

pullover).

- **Low resolution and grayscale.** At 28 x 28 grayscale the model cannot learn colour, fabric or fine construction details, so its outputs have no practical design value.
- **No sample-quality metric.** I evaluated generated images visually and through reconstruction error. Feature-based metrics such as the Fréchet Inception Distance (FID; Heusel et al., 2017) compare the statistics of generated and real images and would make sample quality measurable and comparable.
- **Unused capacity and class mixing.** Only 9 of 16 latent dimensions are used, and shirt, pullover and coat overlap in latent space.

Next steps. (1) Tune β below and above 1 and measure the trade-off more finely (Higgins et al., 2017). (2) Train a conditional VAE that also receives the class label, so that each item can be generated on request and the shirt, pullover and coat confusion is reduced. (3) Try different latent sizes, guided by the active-dimension diagnostic. (4) Add FID as a quantitative sample-quality metric (Heusel et al., 2017). (5) Move to a colour, higher-resolution clothing dataset with a stronger decoder, and keep the memorisation check as a required step.

7. References

Carlini, N., Hayes, J., Nasr, M., Jagielski, M., Sehwag, V., Tramèr, F., Balle, B., Ippolito, D., & Wallace, E. (2023). Extracting training data from diffusion models. In *Proceedings of the 32nd USENIX Security Symposium* (pp. 5253–5270). USENIX Association. https://www.usenix.org/conference/usenixsecurity23/presentation/carlini

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